

TSPI AI Engine

Final Domain-Expert Response to the Development Team

This document provides the complete official response to the remaining questions in “TSPI AI Engine — What We Need From You to Finish the Project.”

The development team reports that the general engine architecture, scoring framework, safety handling, and propose-only learning process have already been built and tested. The remaining work concerns the authoritative clinical data, the corrected Module Registry, the 180-Network crosswalk, and final confirmation of the master registries. □ □

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This response must be interpreted under the following constitutional principle:

The Thirty-Nine Biological Axes are the central diagnostic and Module-matching layer of TSPI.

The One Hundred Eighty Living Biological Networks explain and validate the mechanisms underlying those Axes. Networks do not replace the Axes and must not become the primary Module-selection layer.

1. Canonical TSPI Architecture

The official reasoning architecture is:

Clinical Evidence



Evidence Integration



Three Biological Keys



Nine Restoration Steps



Twelve Biological Domains



Thirty-Nine Biological Axes



One Hundred Eighty Living Biological Networks



Research and Clinical-Outcome Validation



Module Ranking



Physician Approval



Disease Interpretation and Reports

Disease labels remain clinical interpretations. They are not the primary reasoning engine.

The current frozen ontology is:

framework_version = 39-axis-master-260715

The corrected Axis list on pages 36–41 is the canonical list. The older detailed section on pages 42–60 is superseded because it uses the former A35 name and omits A36–A39. The older per-Axis Module lists must not be used directly; they may only be treated as legacy material requiring semantic remapping. The same review also confirmed that CELA is Network #120 under A34, not an Axis. □

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2. Official Twelve Domains and Thirty-Nine Biological Axes

The following codes and names must be treated as immutable production identifiers.

Domain 1 — Immune–Infection Governance

A1 — Systemic Inflammatory Load

A2 — Immune Homeostasis

A3 — Pathogen Defense

A4 — Post-Infectious Persistence

Domain 2 — Metabolic–Energy Core

A5 — Glucose–Insulin Regulation

A6 — Lipid–Lipotoxicity

A7 — Nutrient Sensing

A8 — Mitochondrial Network

A9 — Oxidative Stress & Redox Balance

A10 — Proteostasis–Autophagy–Lysosome

Domain 3 — Genomic Programming

A11 — DNA Repair & Genomic Stability

A12 — Epigenetic Regulation

A13 — Longevity Signaling

A14 — Senescence & SASP

Domain 4 — Detoxification & Cellular Defense

A15 — Detoxification Capacity

Domain 5 — Digestive–Gut Ecosystem

A16 — Digestive–Absorptive Function

A17 — Microbiome Ecology

A18 — Gut Barrier & Mucosal Integrity

A19 — Colon Biofilm & Mucosal Burden

Domain 6 — Vascular–Circulation

A20 — Endothelial Function

A21 — Vascular Remodeling & Microcirculation

A22 — Thrombosis–Hemostasis

A23 — Hemodynamics & Arterial Elasticity

Domain 7 — Repair–Regeneration

A24 — Wound Healing

A25 — Fibrosis & ECM Remodeling

A26 — Stem Cell & Hematopoiesis

A27 — Glycation / Carbonyl Stress

Domain 8 — Musculoskeletal & Structural Integrity

A28 — Muscle Protein Turnover

A29 — Bone Remodeling

A30 — Connective Tissue Integrity

Domain 9 — Endocrine Network

A31 — Female Hormone Network

A32 — Male Androgen Network
A33 — Thyroid–Adrenal–Growth Axis
Domain 10 — Neurological Regulation
A34 — Neuro–Sleep–Stress–Brain Clearance
Domain 11 — Organ Resilience
A35 — Visceral Organ Functional Reserve
A36 — Local Organ & Tissue Interface
Domain 12 — Oncology & System-Level Regulation
A37 — Oncologic Cell Fate
A38 — Tumor Microenvironment & Metastatic Dynamics
A39 — Genomic Regulation Homeostasis

Engineering lock

No AI model may:

rename an Axis;

create an additional Axis;

infer an Axis identity from a disease label;

substitute a legacy Axis number;

or use a Network name as an Axis.

All Axis identities must resolve from the official Axis Registry.

3. Blocker 1 — Approved Module Registry with Current Axis Codes

Official decision

Please choose:

The development team should generate the candidate legacy-to-current remap with conversion types and confidence flags. The TSPI clinical/domain-expert team will then review and approve it.

The development team is authorised to prepare the candidate remap, but it is not authorised to approve clinical mappings automatically.

The current Module Registry cannot enter production because it contains legacy numbering, current numbering, mixed numbering, and deprecated Axis definitions. The prior review found old and new mappings within the same file, including the particularly dangerous case where legacy A27 meant “Metabolic Balance” while current A27 means “Glycation / Carbonyl Stress.” □

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Blind numerical conversion is prohibited

The system must not apply rules such as:

$\text{legacy_axis_number} - 1 = \text{current_axis_number}$

Even when that appears correct for some rows, the legacy file uses the same numbers for different biological meanings.

The correct principle is:

Semantic meaning determines the mapping—not the legacy number.

Required Candidate Module Registry fields

module_code

module_name

phytochrome_code

legacy_axis_number

legacy_axis_name

candidate_current_axis_codes[]
conversion_type
conversion_confidence
mapping_reason
mapping_status

primary_restoration_step
secondary_restoration_steps[]

contraindications[]
drug_herb_interactions[]
dose_type

evidence_status
evidence_grade

axis_master_version
reviewed_by
review_date

Required conversion types

EXACT_SEMANTIC_MAPPING

RENAMED_REFINED

CONTEXT_DEPENDENT

SPLIT_REQUIRES_REVIEW

DEPRECATED_TO_NETWORK

NO_DIRECT_EQUIVALENT

SAME_NUMBER_DIFFERENT_MEANING

Only these may be considered for batch approval after review:

EXACT_SEMANTIC_MAPPING

RENAMED_REFINED

All other types require individual expert review.

Production gate

A Module may participate in clinical Module ranking only when:

mapping_status = APPROVED

axis_master_version = 39-axis-master-260715

production_allowed = true

Before that point, it may be displayed only as:

PROVISIONAL CANDIDATE

NOT FOR CLINICAL USE

“Module matching” must never be interpreted as prescribing.

4. Blocker 2 — Connect the 180 Networks to Axes, Domains, and One Another

The current request states that all 180 Network names have now been verified and loaded, but their Axis codes, Domain assignments, and upstream/downstream connections still require confirmation. □

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The list being loaded does not mean that all biological relationships have been approved.

4.1 Blocker 2a — Confirm the Axis Code for Each Network

Official decision

The development team may populate a best-guess candidate mapping with confidence values, but every row remains provisional until reviewed.

Each Network must contain:

network_code
network_number
network_name_en
network_name_th

primary_axis_code
secondary_axis_codes[]

mapping_reason
mapping_confidence
mapping_status
axis_master_version

Primary and secondary Axes

Each Network must have:

exactly one primary Axis;
zero or more secondary Axes;
an explicit reason for the mapping;
and an approval status.

The primary Axis identifies where the Network principally belongs in the TSPI ontology.

Secondary Axes describe meaningful cross-system effects.

Examples in the Developer sheet

N033 Phase I Detoxification

→ Primary Axis: A15 Detoxification Capacity

N048 Base Excision Repair

→ Primary Axis: A11 DNA Repair & Genomic Stability

N053 DNA Methylation

→ Primary Axis: A12 Epigenetic Regulation

DNA Methylation should not be left without an Axis because it is directly within epigenetic regulation.

Mapping method

Network mapping must be based on:

biological function;
mechanistic scope;
upstream and downstream role;
primary biological consequence;
the official Axis definition.

Network mapping must not be based solely on:

the Network number;
textual similarity;
an organ name;
a disease label;
or an LLM-generated guess.

4.2 Network Code versus Network Number

Official decision

network_code must not be treated as merely the zero-padded network_number.

These fields serve different purposes:

network_number = human-readable display order from 1 to 180

network_code = permanent and stable identity of the Network

The current N001–N180 values may be retained as legacy or display identifiers, but the system should have a permanent primary identifier independent of presentation order.

Example:

network_code = LBN-DET-PHASE1

network_number = 33

legacy_display_id = N033

A future Registry version may insert, merge, split, or reorder Networks. Such changes must not alter the identity of an already approved Network.

Engineering rule

The AI must never derive biological meaning from a Network code or number.

Meaning comes only from the official Network Registry fields.

4.3 Blocker 2b — Confirm the Domain for Each Network

Official decision

The primary Domain should be derived automatically from the confirmed primary Axis.

It should not be maintained as an independent manually authored source of truth.

Use:

primary_domain_code

=

AxisRegistry[primary_axis_code].domain_code

Examples:

N033 → A15 → Domain 4

N048 → A11 → Domain 3

N053 → A12 → Domain 3

When a Network has secondary Axes, the system may also derive:

secondary_domain_codes[]

from those Axes.

Why this is necessary

Automatic derivation prevents:

an A15 Network being entered under the wrong Domain;

a future Axis update leaving stale Network-Domain data;

duplicate editing;

and conflicts between the Axis and Network Registries.

The CONFIRM_domain_num column may be used as a review field, but the final Domain relationship must remain dependent on the approved Axis Registry.

4.4 Blocker 2c — Network Connections

Official decision

Create an Official Network Edge Registry.

A simple upstream/downstream list is insufficient because it does not identify the biological nature of the relationship.

Required fields:

source_network_code

target_network_code

edge_type

direction

biological_effect

context

evidence_status

evidence_grade

source_reference

confidence

mapping_status

version

Recommended edge_type values:

ACTIVATES

INHIBITS

REQUIRES

COMPENSATES

PROPAGATES_TO

RESTORES

FAILURE_PRECEDES

ASSOCIATED_WITH

Each connection must distinguish among:

causal evidence;

mechanistic evidence;

compensatory interaction;

clinical association;

and hypothesis.

The AI may propose a candidate Edge and retrieve supporting evidence, but it may not publish or modify an Official Edge without scientific review.

5. Correct Interpretation of the 25% Network-Match Component

The Developer document states that confirming the Network-Axis/Domain crosswalk will enable the 25% Network Match component.

This must be implemented carefully.

Incorrect architecture

Network

→ Module candidate

Correct TSPI architecture

Thirty-Nine Biological Axes

→ Official Axis-to-Module candidate generation

→ Safety gate

→ Network mechanistic validation

→ Research-evidence validation

→ Similar-case and clinical-outcome validation

→ Module ranking

→ Physician approval

The Network layer is used to:

explain why an Axis is disturbed;

validate whether a Module's mechanisms are relevant;

assess mechanistic coverage;

detect redundant mechanisms;

and support ranking after Axis-based candidates have been generated.

It must not replace Axis matching.

When Network data are incomplete

Do not enter:

network_match = 0

because absence of data does not mean absence of mechanistic correspondence.

Use:

network_validation_status = NOT_ASSESSED

The output should show both the provisional score and how much of the scoring model has actually been validated.

Example:

Provisional Module Match: 82/100

Validated Score Coverage: 75%

Network Validation: NOT_ASSESSED

Overall Module Confidence: REDUCED

Do not silently renormalise missing components.

The earlier Developer document correctly recognised that missing Network data must remain visible rather than being hidden by reweighting. □

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6. Clinical Master Dataset 2.1 — Marker Registry

The proposed Marker Registry is acceptable as a starting point but requires additional fields to support medical-grade reasoning.

Required schema

marker_code

marker_name_en

marker_name_th

specimen_type

measurement_method

unit

value_source_type

calculation_rule

formula_version

required_inputs[]

primary_axis_code

secondary_axis_codes[]

direction_rule

curve_type
context_rules[]
population_rules[]

evidence_status
evidence_grade
source_reference

status
version

6.1 Measured versus Derived Values

Use:

value_source_type:

- MEASURED
- DERIVED

A derived marker is not a direction rule.

Example:

marker_code = HOMA_IR
value_source_type = DERIVED
direction_rule = LOWER_BETTER

The formula must be unit-specific.

When glucose is in mg/dL:

$\text{HOMA-IR} = \text{fasting_glucose} \times \text{fasting_insulin} \div 405$

When glucose is in mmol/L:

$\text{HOMA-IR} = \text{fasting_glucose} \times \text{fasting_insulin} \div 22.5$

The system must validate units before calculation.

6.2 Direction Rules

Use these top-level categories:

LOWER_BETTER

HIGHER_BETTER

TARGET_RANGE

CONTEXT_DEPENDENT

U_SHAPE should not be a separate top-level direction rule.

It should be represented as:

direction_rule = TARGET_RANGE

curve_type = U_SHAPE

The Registry may also support:

curve_type:

- LINEAR
- U_SHAPE
- J_SHAPE
- ASYMMETRIC

6.3 Ferritin Mapping

Ferritin is a context-dependent marker.

The official mapping is:

Primary Axis:

A26 — Stem Cell & Hematopoiesis

Secondary Axis:

A1 — Systemic Inflammatory Load

when an inflammatory context is supported

Direction Rule:

CONTEXT_DEPENDENT

Low ferritin

Low ferritin or depleted iron storage primarily supports A26.

High ferritin with inflammatory evidence

High ferritin combined with CRP, hs-CRP, ESR, inflammatory symptoms, or another inflammatory pattern may support A1.

High ferritin alone

High ferritin alone must not automatically be interpreted as systemic inflammation.

The previous reference to A34 as inflammation was a typographical error. A34 is Neuro–Sleep–Stress–Brain Clearance.

7. Clinical Master Dataset 2.2 — Laboratory Cut-off Registry

The proposed structure should be expanded as follows:

marker_code

unit

measurement_method

population_group

sex

age_min

age_max

pregnancy_status

fasting_status

optimal_low

optimal_high

subclinical_low

subclinical_high

functional_low

functional_high

pathological_low

pathological_high

critical_low

critical_high

source_reference

status

version

No universal global cut-off

Thresholds may differ according to:

sex;
age;
child, adult, or older-adult population;
pregnancy;
fasting status;
assay method;
measurement unit;
comorbid conditions;
and medication effects.

The AI must not invent a cut-off from memory or from general language-model knowledge.

Disturbance bands

Use:

OPTIMAL

SUBCLINICAL

FUNCTIONAL_DISTURBANCE

PATHOLOGICAL

CRITICAL

A Critical threshold triggers a safety action. It does not automatically become the Axis severity score.

8. Clinical Master Dataset 2.3 — Evidence Grade Registry

Official decision

Do not assign a single Evidence Grade to an entire Module.

Evidence Grade must be evaluated at the level of:

Module + Claim + Target Axis/Network + Evidence Source

Example:

Module X

Claim: reduces IL-6

Target: A1

Grade: B

Module X

Claim: increases ATP production

Target: A8

Grade: C

Required schema

module_code

claim_code

claim_text

target_axis_code

target_network_codes[]

research_id

study_type

sample_size

comparator

effect_size

quality_score

evidence_grade
verified_by

status
version

Evidence Grade definitions

Grade A

High-quality and directly relevant human evidence, such as strong controlled clinical evidence, replicated clinical trials, or equivalent high-level evidence for the specific Module, claim, and target.

Grade B

Relevant human pilot, controlled clinical, or observational evidence with limitations.

Grade C

Animal, cell, molecular, omics, or mechanistic evidence.

Grade D

Traditional knowledge, expert opinion, indirect evidence, or evidence that has not yet been experimentally confirmed.

Ungraded evidence

Do not default every unreviewed Module to Grade D.

Use:

EVIDENCE_NOT_GRADED

“Not yet graded” is not the same as “weakest evidence.”

The Developer report currently notes that all Modules default to D because the Registry lacks grades; this should be corrected. □

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9. Clinical Master Dataset 2.4 — Module to Nine Restoration Steps

Each Module may have:

one primary Restoration Step;
multiple secondary Restoration Steps;
and evidence supporting every Step mapping.

Official Nine Restoration Steps

Detoxification

Microbiome Restoration

Immune–Inflammatory Regulation

Redox Homeostasis

Metabolic and Mitochondrial Restoration

Autophagy and Cellular Clearance

Genomic Stability

Genomic Regulation Homeostasis

Prakati

These are constitutional TSPI Steps. Their names and meanings must not be replaced by diseases, organs, marketing claims, or clinical subgoals.

For example, these are not valid replacements:

Step 3 = Diabetes

Step 4 = Liver

Step 7 = Sleep

Diabetes, liver dysfunction, sleep disturbance, glycation, and lipid metabolism may be targets or mechanisms under the architecture, but they are not replacements for the official Nine Steps.

Required schema

module_code

primary_step_code

secondary_step_codes[]

step_claims[]

evidence_grade

mapping_status

version

10. Clinical Master Dataset 2.5 — Per-Axis Minimum Evidence

A simple Low/Medium/High field may be retained as a human-readable summary, but it is not sufficient for the engine.

Required schema

axis_code

core_evidence[]

supporting_evidence[]

optional_evidence[]

minimum_core_count

minimum_total_weight

required_modalities[]

accepted_inputs[]

exclusion_rules[]

allow_provisional_assessment

status

version

Evidence weights

Use the previously agreed structure:

CORE = 3

SUPPORTING = 2

OPTIONAL = 1

The Developer review already acknowledged that completeness must be weighted and that a missing CORE marker cannot be treated as “almost complete.” □

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Axis assessment states

NOT_ASSESSED

AXIS_CANDIDATE

PROVISIONALLY_ASSESSED

ASSESSED

HIGH_CONFIDENCE_ASSESSED

General rules

Symptoms may create an Axis candidate.

A hypothesis does not create a final score.

An Axis must satisfy its minimum-evidence gate before scoring.

Missing evidence must never be entered as normal.

Different Axes require different evidence standards.

Examples

A26 — Stem Cell & Hematopoiesis

At minimum, a CBC or equivalent haematological evidence is required.

A deeper conclusion may require:

reticulocyte count;

ferritin and iron studies;

vitamin B12 and folate;

haemoglobin typing;

or bone-marrow evidence depending on context.

A8 — Mitochondrial Network

Fatigue or a metabolic-energy phenotype may create an Axis candidate.

A high-confidence mitochondrial conclusion requires objective supporting evidence.

A37 — Oncologic Cell Fate

Requires verified oncology evidence, such as:

pathology;

imaging;

physician-confirmed cancer diagnosis;

or validated molecular/oncologic evidence.

General symptoms alone must never generate an A37 score.

Implementation decision

The development team may prepare a candidate Minimum-Evidence Registry for all 39 Axes based on the official Axis definitions.

Every row must remain:

status = CANDIDATE_FOR_EXPERT_REVIEW

until approved.

11. Section 3 — Content to Ratify

The provisional content was seeded from worked examples and is currently marked non-authoritative. That is the correct interim status. □

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11.1 Red-Flag List

Decision

Approve the architecture, but do not yet approve every seeded rule as authoritative.

Retain:

authoritative = false

status = CANDIDATE

Each Red-Flag Rule must include:

trigger

population

unit

clinical_context

urgency

required_action

override_behavior

source_reference

version

The Red-Flag Engine must run before:

Axis reasoning;
Network reasoning;
Module matching;
and report generation.
Red flags must remain visible even when a summary score is low.

11.2 Symptom Dictionary

Decision

Approve the structure, but every symptom-to-phenotype and phenotype-to-Axis mapping must be reviewed before becoming authoritative.

The correct logic is:

Single symptom

→ nonspecific evidence

Coherent symptom cluster

→ named phenotype

Named phenotype

→ weighted Axis evidence

Example

Bloating alone

→ nonspecific

Bloating + early satiety + belching

→ Upper Digestive Dysfunction Phenotype

→ supports A16

Constipation clarification

Constipation may support:

A17 when Microbiome/Dysbiosis context is present;

A19 when evidence specifically supports colon biofilm or mucosal burden.

Constipation alone must not confirm A19.

Stress-related worsening

Symptoms worsening with stress may support:

A34;

and CELA Network #120.

It must not independently confirm either one.

11.3 Negative-Test Rules

Decision

Approve the concept, but each rule must be specific to the validated scope of the test.

A negative result reduces only the probability of abnormalities that the test can actually detect.

Examples:

A normal gastroscopy does not establish that A16, A17, and A18 are normal.

A normal brain MRI does not establish that A34 is normal.

Normal routine blood tests do not establish that all 39 Axes are normal.

A negative test may change:

probability

confidence

differential ranking

It must not automatically set:

Axis = NORMAL

11.4 Critical Thresholds

The seeded thresholds may be used only as provisional safety alerts.

They must not yet become universal official clinical thresholds.

This includes values such as:

potassium below 2.5 or above 6.5;

sodium below 120 or above 160;

glucose below 54 or above 450;

haemoglobin below 7;

platelets below 20;

AST or ALT above 1,000;

oxygen saturation below 88%;

troponin above 0.04.

Thresholds must be linked to:

units;

assay method;

population;

age;

pregnancy status;

comorbidities;

and clinical context.

Troponin in particular must use the assay-specific upper reference limit, not one universal value.

Retain:

authoritative = false

validation_status = PROVISIONAL

12. Section 4 — Phenotype Registry

Official decision

Confirm Option A:

A Phenotype is a named, structured, and version-controlled clinical pattern.

The development team should propose the first Phenotype Registry.

Recommended initial scope

Approximately 30 Core Phenotypes for Version 1

This is an initial reviewable scope, not a permanent maximum.

All proposed Phenotypes must begin as:

status = CANDIDATE

Required fields

phenotype_code

phenotype_name_th

phenotype_name_en

required_features[]

supporting_features[]

exclusion_features[]

red_flag_overrides[]

candidate_axis_codes[]
axis_evidence_weights[]

minimum_feature_count
confidence_rule

status
version
approved_by
Correct architecture
Symptoms + Signs
→ Named Phenotype
→ Axis Evidence

Phenotype is one evidence route, not the entire TSPI reasoning engine.

Laboratory, imaging, omics, medication, exposure, physiology, and clinical history must enter Evidence Integration through their respective Registries.

13. Section 5.1 — A35 and A36 Organ Assignment

A35 — Visceral Organ Functional Reserve

A35 evaluates:

The functional capacity and reserve of a whole internal organ under biological stress.

It may include whole-organ functional reserve relating to:

liver;
kidney;
heart;
lung;
pancreas;
gallbladder and biliary organs;
spleen;
gastrointestinal organs;
bladder;
and other visceral organs.

Relevant evidence may include:

organ-function tests;
clearance or filtration;
synthetic function;
perfusion;
imaging;
functional-reserve testing;
and longitudinal recovery.

A35 does not replace mechanism-specific Axes such as:

A15 Detoxification Capacity;
A20 Endothelial Function;
A25 Fibrosis & ECM Remodeling;
or A33 Thyroid–Adrenal–Growth Axis.

It is a whole-organ reserve layer.

A36 — Local Organ & Tissue Interface

A36 evaluates:

A local tissue, mucosal surface, sensory-organ site, or organ–environment interface.

Examples include:

eye, retina, cornea, and related local ocular tissue;

ear and hearing apparatus;

nose, sinus, throat, and local ENT tissue;

oral mucosa;

local skin or mucosal interfaces;

reproductive and pelvic tissues;

prostate or local genital tissue;

anorectal tissue;

local urinary mucosa.

Relevant evidence may include:

local examination;

endoscopy;

audiometry;

OCT;

local imaging;

cytology or pathology;

local microbiology;

tissue-specific functional testing.

The distinction is:

A35 = whole-organ functional reserve

A36 = local tissue, mucosal, or organ-interface state

14. Section 5.2 — Historical Response Data

Official decision

Choose:

Stay neutral for now.

When standardised historical Outcome data are absent, do not enter:

historical_response = 0

Use:

historical_response_status = NOT_ASSESSED

Absence of historical evidence does not mean failure.

Historical response may influence ranking only when the data include:

baseline status;

Axis profile;

Module and dose;

treatment duration;

follow-up;

defined outcome;

adherence;

adverse events;

confounders;

data completeness;

physician validation.

Population learning

Use the following interim rule:

$n < 30$

→ descriptive only

$n \geq 30$

→ candidate population signal

→ propose-only

→ clinician review required

The system must never auto-update the global model from a single patient.

The earlier Developer review correctly identified that population learning must remain propose-only and require clinician approval. □

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15. Section 5.3 — Report Language and Content

Official decision

Choose:

Thai-first bilingual reporting.

Physician Report

The Physician Report should include:

Thai explanation with canonical English terminology;

Axis, Network, and Module codes;

Evidence Chain;

Evidence Status;

Evidence Grade;

Confidence;

Data Completeness;

Biological Uncertainty;

research references;

similar-case evidence;

reason_selected;

reason_not_selected;

Registry and calculation versions.

Patient Report

The Patient Report should:

use Thai as the primary language;

include English technical terminology only where useful;

use the same Analysis Object as the Physician Report;

preserve the same scientific conclusions;

include Red Flags and a monitoring plan;

and require physician approval before release.

Multi-Omics Report

The Multi-Omics Report must:

separate measured Omics from phenotype-based inference;

use NOT_AVAILABLE when data do not exist;

avoid implying that every patient requires all Omics tests;

recommend testing according to clinical utility and value of information.

16. Section 5.4 — Additional Missing Datasets and Controls

The following should be added to the project plan:

Axis Registry version lock

Network Edge Registry
Imaging Evidence Registry
Medication and Exposure Registry
Research Claim Registry
Clinical Outcome Definition Registry
Drug–Herb Interaction Evidence Registry
Unit and Assay Normalisation Registry
Data Provenance and Source IDs
Patient Identity and Data-Integrity Gate
Temporal and Longitudinal Patient Model
Reasoning Ledger
Contradiction Engine
Cross-Report Consistency Validator
Biological Truth Validation Layer
Physician Electronic Approval
Separation of Module, Product, and Phytocore entities
Entity-count reconciliation

The system must clarify the difference among:

145 pre-listed entries in the template;
200 Biological Modules;
242 Product Codes.

These must not be treated as the same entity.

The ontology should distinguish:

Biological Module
Commercial Product / H-Code
Phytocore

17. Recommended Completion Order

Freeze the Official Axis Master.

Generate the Candidate Module Remap.

Review and approve the Module Registry.

Review all 180 rows in network_crosswalk_FULL_DRAFT.csv.

Confirm each Network's primary and secondary Axes.

Derive Domain assignments automatically from approved Axes.

Build and approve the Network Edge Registry.

Complete the Marker Registry.

Complete the Laboratory Cut-off Registry.

Grade evidence at claim-and-target level.

Map Modules to the Nine Restoration Steps.

Complete Minimum Evidence rules for all 39 Axes.

Review Red Flags, Symptoms, Negative Tests, and Critical Thresholds.

Review the first 30 Core Phenotypes.

Run deterministic validation and regression tests.

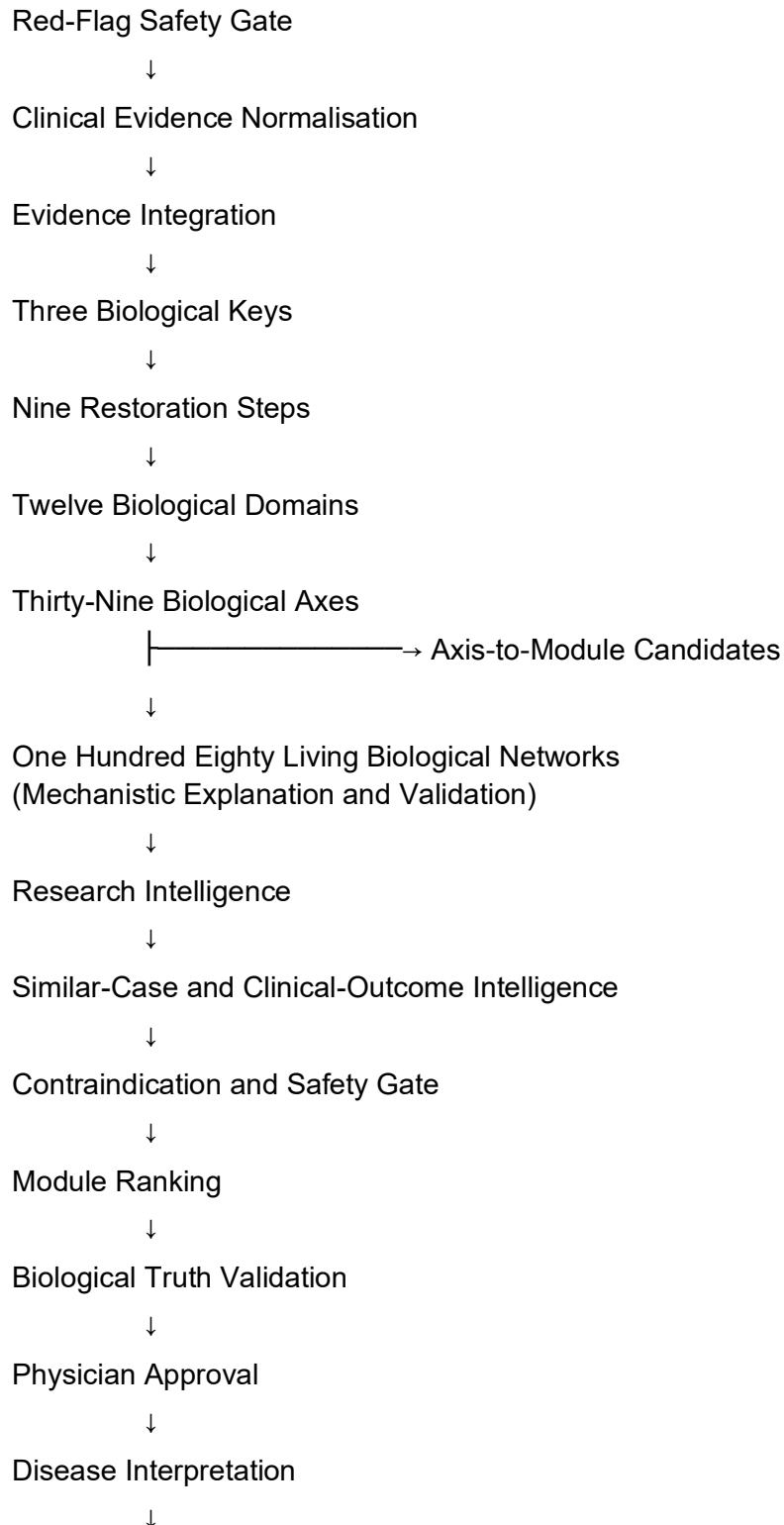
Begin a physician-only pilot.

Release patient-facing reports only after validation.

18. Required Final TSPI AI Pipeline

Patient Identity and Data-Integrity Gate

↓



Physician, Patient, and Multi-Omics Reports

Final Developer Lock

TSPI AI must be Registry-driven, not LLM-driven.

The Thirty-Nine Biological Axes are the centre of biological assessment and Module-candidate generation.

The One Hundred Eighty Living Biological Networks provide mechanistic explanation and validation; they do not replace the Axes.

Missing data must produce NOT_ASSESSED, never zero and never a default score of 50.

Safety is a gate, not a numerical score.

A hypothesis must not generate an authoritative Axis score.

The LLM must never invent an Axis score, Network identity, Module identity, research result, clinical case, or treatment outcome.

Every recommendation must be traceable to patient evidence, an approved Registry entry, a Research ID, and—when used—validated clinical-outcome data.

Module matching is not prescribing. All clinical plans require physician approval.

The earlier Developer implementation already reports several correct safeguards: a deterministic engine, no LLM assignment of Axis scores or Network identities, official-Registry-only Module resolution, no default score of 50, explicit NOT_ASSESSED, separate Evidence Status and Evidence Grade, weighted Data Completeness, Red-Flag-first screening, Safety as a gate, propose-only population learning, and version stamping. These controls must remain permanent regression requirements and must never be removed in later releases. □

TSPI_Open_Questions_And_Blockers_For_Experts_260716_143304 (1).pdf

Critical TSPI Developer Directives

Non-Negotiable Rules to Prevent Architectural Drift

The following points should be treated as hard engineering constraints, not optional recommendations. They are the areas most likely to cause TSPI to drift away from its intended philosophy or produce clinically unsafe output.

1. The Thirty-Nine Biological Axes Are the Central Reasoning Layer

This is the single most important rule.

The Thirty-Nine Biological Axes are the official representation of the patient's Biological State. They are the central layer that connects Clinical Evidence to biological interpretation, Network explanation, and Module matching.

The system must not replace the Axes with:

disease labels;

symptoms;

laboratory markers;

the 180 Networks;

Modules;

or LLM-generated summaries.

The correct principle is:

Clinical Evidence

→ Evidence Integration

→ Thirty-Nine Biological Axes

→ Network Explanation

→ Module Candidates

→ Clinical Interpretation

The Three Keys, Nine Restoration Steps, and Twelve Domains provide higher-level organization and interpretation, but the Thirty-Nine Axes remain the central diagnostic and Module-matching layer.

Developer acceptance rule

Every Module recommendation must be able to answer:
Which approved Axis or Axes generated this Module candidate?
If this cannot be answered, the Module must not be recommended.

2. Disease Labels Must Never Drive the Biological Analysis

TSPI is not:

Disease

→ Module

It is also not:

Disease

→ Network

→ Module

The correct process is:

Clinical Evidence

→ Biological State

→ Module Restoration Plan

→ Disease Interpretation

Disease labels may be used for:

clinical communication;

differential diagnosis;

ICD coding;

guideline comparison;

insurance documentation;

and final medical interpretation.

They must not become the source of Axis scores or Module selection.

Required regression test

Remove the disease name from a test case while keeping the history, symptoms, laboratory results, imaging, and other evidence.

The Biological Axes should remain substantially unchanged.

If removing the disease label materially changes the Axis analysis, the engine is still disease-driven and must be corrected.

3. No LLM May Generate Clinical Truth

The LLM is a language and explanation layer. It is not the source of biological authority.

The LLM must never independently generate:

an Axis identity;

an Axis score;

a Network identity;

a Network connection;

a Module name;

a Module–Axis mapping;

a research finding;

a clinical case;

a response rate;

a recovery time;

an interaction result;

or a contraindication.

All such information must come from approved structured data.

The LLM may:

summarize;

explain;

translate;

organize;

and communicate validated reasoning.

It must not create the reasoning inputs or authoritative results.

Structured reasoning must exist before narrative generation. The report is a translation of the Reasoning Object, not the source of the reasoning.

4. There Must Be One Authoritative Analysis Object Per Patient Run

The Physician Report, Patient Report, and Multi-Omics Report must all read from the same immutable Analysis Object.

They must not calculate their own:

Three-Key scores;

Axis scores;

confidence values;

Module rankings;

evidence statuses;

or clinical priorities.

The same analysis_run_id must be present in every report.

Example:

analysis_run_id

patient_data_version

axis_registry_version

network_registry_version

module_registry_version

marker_registry_version

cutoff_registry_version

calculation_version

report_template_version

Hard consistency rule

If the Physician Report says:

A27 = 82

the Patient and Multi-Omics reports must show the same value, status, evidence basis, and confidence.

The wording may differ. The scientific conclusion must not.

A Cross-Report Consistency Validator must run before release.

5. Missing Data Must Never Become Normal, Zero, or Default 50

This rule must remain permanent.

No evidence ≠ normal

No evidence ≠ zero disturbance

No evidence ≠ score 50

The correct output is:

NOT_ASSESSED

or, where limited evidence exists:

AXIS_CANDIDATE

PROVISIONALLY_ASSESSED

A numerical score may be produced only after the Axis satisfies its approved minimum-evidence gate.

Prohibited behavior

Missing data → 50/100

Missing marker → severity 0

No Network data → Network Match 0

No historical outcomes → Historical Response 0

Correct behavior

Missing data → NOT_ASSESSED

No Network data → NETWORK_VALIDATION_NOT_ASSESSED

No historical outcomes → HISTORICAL_RESPONSE_NOT_ASSESSED

The system must reduce Confidence or score coverage, not falsely reduce biological severity.

The prior Developer review correctly established explicit NOT_ASSESSED, no default 50, weighted completeness, and the rule that missing CORE evidence must not appear nearly complete. These safeguards must never be removed.

6. Evidence Status, Evidence Grade, Confidence, and Completeness Are Different Variables

These concepts must not be merged.

Evidence Status

What kind of statement is this?

MEASURED

DERIVED

CLINICAL_INFERENCE

HYPOTHESIS

NOT_AVAILABLE

NOT_ASSESSED

Evidence Grade

How strong is the supporting scientific evidence?

A

B

C

D

EVIDENCE_NOT_GRADED

Confidence

How confident is the engine in the current interpretation?

Data Completeness

How much of the expected evidence for that Axis is actually present?

These fields answer different questions.

For example:

HbA1c = MEASURED

HOMA-IR = DERIVED

A5 score = DERIVED

Mitochondrial dysfunction = CLINICAL_INFERENCE

Direct mitochondrial failure = NOT_ASSESSED

Do not label an Axis score as MEASURED merely because it was calculated from measured markers.

7. Confidence Must Never Change Biological Severity

A severe abnormality with incomplete evidence remains severe, but uncertain.

It must not become biologically mild because Confidence or Completeness is low.

Correct interpretation:

Axis severity: severe

Axis confidence: low

Data completeness: limited

Biological uncertainty: high

Incorrect interpretation:

Severe × low confidence × low completeness

= mild final severity

Data quality affects the reliability of the assessment, not the biological direction of the available evidence.

This distinction is essential for NSS and every Axis score.

8. Safety Is a Gate, Not a Score

Do not use:

Safety Score = 94/100

because it suggests that an unsafe Module may remain eligible if its total score is high enough.

Use categorical safety outcomes:

PASS

PASS_WITH_MONITORING

HOLD

CONTRAINDICATED

INSUFFICIENT_SAFETY_DATA

The processing order must be:

Module Candidate

→ Contraindication Check

→ Drug–Herb Interaction Check

→ Organ-Function Restriction

→ Dose Restriction

→ Monitoring Requirements

→ Safety Decision

If the Module fails the Safety Gate, it must be excluded before ranking.

The earlier Developer implementation correctly adopted Safety as an exclusion gate. That rule must remain immutable.

9. Networks Explain and Validate; They Do Not Replace Axis Matching

The 180 Living Biological Networks must not become a second competing diagnostic system.

Their role is to explain:

why an Axis is disturbed;

how disturbances propagate;
what compensatory mechanisms are active;
how different Axes communicate;
how a Module may restore the mechanism;
and whether Module mechanisms overlap or complement one another.

The correct sequence is:

Axis State

→ Axis-to-Module Candidate Generation

→ Safety Gate

→ Network Mechanistic Validation

→ Research Validation

→ Clinical-Outcome Validation

→ Ranking

Not:

Network

→ Module

The 25% Network component must therefore be treated as mechanistic validation coverage, not as the primary candidate-generation mechanism.

Until the official Network Registry is approved:

network_validation_status = NOT_ASSESSED

Do not enter zero. Do not silently renormalize.

10. Network Codes Must Be Stable Identities, Not Biological Meaning

Keep separate fields:

network_code

network_number

legacy_display_id

network_number is display order.

network_code is the permanent reference used by APIs, graph edges, databases, and reports.

The engine must never infer biological meaning from either code or number.

All meaning must come from:

approved name;

definition;

Primary Axis;

Secondary Axes;

scope;

upstream/downstream relationships;

evidence;

and version.

If a Network is split, merged, renamed, or reordered, the history must remain traceable.

11. Domain Assignment Should Be Derived From Axis Assignment

Do not manually maintain the same biological relationship in multiple places.

Once a Network's Primary Axis is approved:

Network Primary Domain

=

Axis Registry Domain of Primary Axis

The same applies to Secondary Domains.

This eliminates duplicate truth sources and prevents contradictions between:

Network Registry;

Axis Registry;

and Domain Registry.

Manual confirmation may be used during review, but the final production relationship must be derived.

12. Official Registries Must Be the Only Source of Clinical Authority

Each knowledge type must have one owner.

Examples:

Axis definitions → Axis Registry

Network definitions → Network Registry

Module identities → Module Registry

Marker interpretation → Marker Registry

Cut-offs → Laboratory Cut-off Registry

Contraindications → Module Safety Records

Research claims → Research Claim Registry

Clinical outcomes → Clinical Outcome Registry

The same fact must not be manually authored in multiple places.

Generated tables may exist as views, but they must not become separate editable sources.

13. Research Intelligence Must Be Traceable

The engine must never display precise scientific claims without a source record.

Do not display claims such as:

Reduced ALT by 42%

Response rate 89%

Median recovery 31 days

unless the report can link them to an approved record containing:

research_id

source_document_id

study_type

sample_size

population

intervention

comparator

dose

duration

outcome_definition

effect_size

statistical_result

quality_score

evidence_grade

verified_by

A Module does not have one universal Evidence Grade.

The Grade must be attached to:

Module + Claim + Target + Evidence Source

No source ID means no precise quantitative claim.

14. Clinical Case Intelligence Must Separate Individual Cases From Cohort Evidence

A single case cannot have a “response rate.”

The system must distinguish:

Nearest individual cases

case_id

similarity_score

similarity_components

treatment

timeline

individual_outcome

Matched cohort results

cohort_definition

cohort_size

similarity_threshold

responder_definition

responders / total

median_response_time

adverse_events

confounders

data_completeness

Do not generate synthetic case numbers, response rates, or recovery periods.

Every displayed case must be retrievable from the Clinical Outcome Database and appropriately de-identified.

15. Similarity Must Be Biological, Not Disease-Label Similarity

Case matching should consider:

Thirty-Nine Axis profile;

Axis Confidence;

age and sex;

symptom and phenotype profile;

laboratory patterns;

imaging;

disease stage;

medication exposure;

organ function;

lifestyle and environment;

prior treatment;

and data completeness.

Disease name alone must not dominate the similarity score.

The system must show which variables contributed to the similarity result.

16. Negative Tests Must Be Limited to Their Actual Scope

A negative test does not prove the entire biological system is normal.

Examples:

Normal gastroscopy

≠ normal digestion, microbiome, and gut barrier

Normal brain MRI

≠ normal sleep, stress, autonomic function, or brain clearance

Normal routine blood tests

≠ all 39 Axes normal

A negative test may:

reduce probability;

lower Confidence in a hypothesis;

or reorder differential interpretations.

It must not automatically set an Axis to normal.

17. Critical Thresholds Must Be Context- and Assay-Specific

Critical thresholds must include:

unit;

assay method;

patient population;

age;

pregnancy status;

clinical context;

urgency;

and required action.

Do not use one universal troponin threshold. Use the assay-specific upper reference limit.

Provisional thresholds may trigger an alert during development, but they must remain:

authoritative = false

validation_status = PROVISIONAL

until formally reviewed.

18. The Phenotype Layer Must Not Become Another Disease-Classification Layer

Phenotypes are named clinical evidence patterns.

They are not diseases and not final diagnoses.

Correct use:

Symptoms and signs

→ Named Phenotype

→ Weighted Axis evidence

Incorrect use:

Symptoms

→ Phenotype label

→ fixed Axis diagnosis

Each Phenotype must include:

required features;

supporting features;

exclusion features;

Red-Flag overrides;

candidate Axes;

weights;

minimum feature count;

Confidence rules;
version;
approval status.

The Developer team may propose Candidate Phenotypes, but only experts may approve them for clinical authority.

19. The Nine Restoration Steps Must Never Be Renamed Dynamically

The Nine Steps are constitutional TSPI architecture.

They must not be replaced with:

disease names;
organs;
symptoms;
marketing claims;
or ad hoc treatment goals.

A Module may map to:

one Primary Step;
several Secondary Steps.

But every Step assignment requires supporting evidence and version control.

20. Three-Key Scores Must Not Be Invented

The Three Keys are high-level biological dimensions.

Until their scoring formulas are clinically validated, the system should preferentially display:

descriptive state;
supporting Axes;
Confidence;
and Data Completeness.

Do not allow the LLM to produce arbitrary Key scores such as 35/100 or 62/100.

If numerical Key scores are used, they must come from a locked deterministic formula and remain identical across all reports.

21. Reports Must Never Claim Physician Approval Before Approval Exists

Before electronic review and signature, every report must display:

AI-GENERATED CLINICAL DRAFT
PENDING PHYSICIAN REVIEW
NOT FOR PATIENT RELEASE

Only after the physician approves the exact Analysis Run may the status change to:

PHYSICIAN REVIEWED AND APPROVED

Approval must record:

physician identity;
date/time;
report version;
Analysis Run ID;
changes made;
and electronic signature.

22. Every Recommendation Must Include Why It Was and Was Not Selected

For each candidate Module, provide:

reason_selected[]
reason_not_selected[]
safety_status
missing_evidence[]
axis_coverage[]

network_validation[]

research_support[]

clinical_outcome_support[]

This prevents the system from producing an unexplained ranked list and allows clinicians to challenge the reasoning.

23. Biological State Is Longitudinal, Not a Single Snapshot

TSPI must preserve the patient timeline.

Every Axis assessment should include:

baseline

current_state

trend

rate_of_change

intervention_period

response

recovery_or_deterioration

The system must distinguish:

current biological state;

historical state;

acute disturbance;

chronic disturbance;

treatment response;

and predicted trajectory.

Do not overwrite previous Axis states.

24. Learning Must Remain Propose-Only

Machine learning may identify:

candidate marker weights;

candidate Phenotypes;

candidate Module-response patterns;

candidate Network relationships;

and candidate outcome predictors.

It may not directly change:

official Axis definitions;

Module mappings;

safety rules;

Network identities;

evidence grades;

or production scoring.

All changes must follow:

Candidate finding

→ validation

→ expert review

→ versioned approval

→ production release

25. Fail Closed, Not Open

When essential registries are missing, contradictory, or unapproved, the system must stop the affected output.

Examples:

Axis Registry mismatch

→ stop report

Module mapping not approved

→ no clinical Module recommendation

Network identity unresolved

→ Network NOT_ASSESSED

Safety data insufficient

→ Module held

Patient identity mismatch

→ stop all clinical reasoning

The system must never “do its best” by inventing or approximating clinical authority.

Mandatory Pre-Production Acceptance Tests

The system should not enter patient-facing production until all of the following pass:

The same input produces the same structured result under the same Registry versions.

All reports read from the same Analysis Run.

Missing data never become normal, zero, or 50.

No Axis name outside the Official Registry can appear.

No unapproved Module mapping can generate a clinical recommendation.

No unknown Network can receive a score.

No research effect size appears without a Research ID.

No case outcome appears without a Clinical Case or Cohort ID.

Safety failure always excludes the Module.

A hypothesis never produces an authoritative score.

Removing disease labels does not materially change the Biological State.

Negative tests affect only their validated scope.

Patient identity contradiction stops the analysis.

Physician, Patient, and Omics reports are scientifically identical.

No patient report can be released without physician approval.

Every output carries all Registry and calculation versions.

Every Module has traceable reason_selected and reason_not_selected.

Every numerical score has a deterministic calculation record.

Every change to official knowledge has an audit trail.

Regression tests verify that all constitutional rules remain active.

Final Message to the Development Team

Please do not optimise for producing a complete-looking report. Optimise for producing a biologically valid, traceable, and honest result.

A shorter report containing verified information is more valuable than a comprehensive report containing inferred or invented certainty.

TSPI must never hide missing data, unresolved mappings, conflicting evidence, or weak scientific support. Uncertainty is a valid and necessary clinical output.

The platform must remain Axis-centred, Network-explained, evidence-grounded, safety-gated, outcome-informed, version-controlled, auditable, and physician-approved.

The success of TSPI is not measured by how many answers the AI can generate. It is measured by whether every answer can be traced back to the patient's evidence and the approved TSPI biological architecture.